Chapter 4

Bias and Fairness in AI Systems

4.1 INTRODUCTION

As discussed earlier many AI Programs have beaten the best of humans in the business. Twenty-five years back ‘Deep Blue’ defeating Garry Kasparov or AlphaGo beating Lee Sedol in the game of GO have certainly caught our attention. This is sure to evoke mixed feelings, one that is of awe and bewilderment, another may be of fear (AI taking over the human race, singularity) and one another may be that of scepticism. Well, the breakthroughs of AI programs are not only limited to games but also to education, healthcare, transportation, agriculture and many other sectors. As this AI system is getting integrated with many daily activities, there are a few non-tangible issues that are cropping up like Ethics in AI, Fairness and Bias in AI and so on.

Before, we jump into AI of Ethics, let’s revisit some general concepts of ethics. Ethics is a branch of philosophy that deals with moral principles, values and general rules of individual or group behaviour. Basically, this involves the study and exploration of what is considered right and wrong, good and bad, and the moral implications of human actions. Ethics provides a framework for making decisions and determining the appropriateness of behaviour within various contexts. Ethics is of course very different from legal requirements. There are formal rules embodied in terms of law. If there is a deviation from observing the law, it is enforceable and one may face consequences like fine, imprisonment, etc. In contrast, ethics is more like a moral compass, one is not going to face any legal consequence. His or her image or reputation in society may be affected at most. There are several such things which are not unethical but not legal. For example:

- Many companies may be knowingly selling products that can be harmful to consumers or the environment, ranging from addictive social media algorithms and products designed for planned obsolescence (Deliberately using techniques to shorten life span) to traditional harmful products like cigarettes.
- Taking credit for someone else's work or ideas. Plagiarism, even in casual settings, can be a form of intellectual theft and undermines the originality and creativity of others.
- Creating and spreading misinformation, even if not done with malicious intent.

Now let's first understand, what is ethics in designing a software product or generic software development? Ethics in IT refers to the principles and standards of conduct that guide professionals in the Information Technology (IT) industry. It involves making decisions and taking actions that are morally and socially responsible. Key aspects of IT ethics include respecting privacy, protecting data security, ensuring fair use of technology and considering the societal impact of IT advancements. Adhering to ethical principles helps build trust, maintain integrity and promote responsible behaviour within the IT community.

While AI Algorithms have matured significantly in the last decade, algorithms (one of the deliverables of traditional IT) have been present for much longer times. A quick recap about algorithms is, they are a sequence of instructions to complete a task. For any given task, there are many given algorithms at our disposal. We choose the optimal ones, based on how efficient they are in terms of computational time, memory usage, etc. We cannot overlook the fact that machine learning or AI algorithms have been designed differently from conventional IT Programs.

The below table gives a brief summary of traditional software *vs* AI-based software.

Feature	Traditional Software	AI Software
Development	Rule-based, pre-programmed	Data-driven learning from data
Learning	Limited, requires manual updates	Continuous, self-improves with new data
Adaptability	Low, inflexible to changes	High, adjusts to new situations
Generalizability	Limited to specific tasks	High, applies to broader problems
Explainability	Easy to understand, logic-driven	Often a 'black box,' difficult to explain decisions

Let's now explore, some of the issues concerning AI-based software, model or algorithm in more details. Some of the issues are as follows:

1. AI algorithms deal with much more amount of data than traditional algorithms. Because of the amount of data required, often data requires to be collected from scratch. This data may introduce **bias** and hence may question the **fairness** practice adopted in the design and development of the algorithm/model/system (detailed examples are discussed later). This also pronounces the existing issue of **data security and privacy.** Let's look at a couple of issues related to data privacy.
    1. **Informed Consent and Data Collection:** One major concern involves the collection of personal data without clear, informed consent from individuals. AI systems often rely on vast datasets to train and operate effectively. Ensuring that individuals understand how their data will be used and obtaining explicit consent is essential to respect privacy.
    2. **Surveillance and Individual Privacy:** AI technologies, particularly in surveillance systems, can raise concerns about the infringement of individual privacy. Striking a balance between the benefits of enhanced security and the protection of individual privacy rights is a complex challenge that requires careful regulation and oversight.

    Any AI model needs a lot of training data and we the users are providing all this data based on our searches, actions, clicks, likes, comments, etc. Many applications ask us to share location and allow access to microphone, camera, etc. While these come with standard disclaimers, hardly users go through them line by line. As a result, many personal data is available with these applications. On one side, they may be abused by the organizations directly, on the other side, if this data is not safely kept, that may be abused by some third party.

2. The working of the algorithm is not completely known to the designer. Because AI Algorithms are not explicit. For some of the algorithms, the internal working is somewhat understandable and is classified as white box, glass box, etc. The developer or the designer knows how the model parameters are to be determined to best fit the data. However, that knowledge does not make the algorithms completely **transparent**. Hence machine learning algorithms at best can be thought of as meta-algorithms. An Algorithm that can write an algorithm. Here comes a dilemma. In a traditional IT scenario, if someone writes malicious code, the developer will be held accountable. Based on the nature of the maliciousness, it can be legally punishable too. However, in case of AI algorithms in spite of applying sound design principles, the algorithms may produce unwanted results. In this case, who would take the **accountability**? Hence, AI algorithm increases the risk of not being transparent. As a result, **explainability**, **Interpretability**, and **causality** for AI Models have become a very active research area. On the other hand, some models can produce superlative results but are black box models. In this case, how one will ascertain that the outcomes are **trustworthy**?
3. Often, the models are published and made available to the community. Later, we will discuss that even the models expose some risks.

Let's explore some examples. Self-driving cars are an excellent manifestation of AI. It has been tested over many scenarios, road conditions, etc., however, let's say it detects a person lying wounded on the side of the road, will it stop to call the ambulance or will it continue driving? Another example may be, if it sees an animal in the road suddenly appearing, will it break sharply, which may overturn the car and damage itself or will it hit the animal?

Let's take an example of AI System being used in the US Judiciary. Correctional Offender Management Profiling for Alternative Sanctions or COMPAS is a tool, which is used to find the probability/likelihood (can be used equivalently in a loose sense) of an offender becoming a repeat offender (recidivism). This is also applicable to substance usage. Suppose the algorithm had more false positives (i.e., it predicted for a repeat offence, but in reality there was no repeat offence) was high for coloured

individuals. There are many such infamous examples, **Figure 4.1** is an example where a person with criminal history but white skin was treated as low risk compared to a first-time offender with black skin.



Figure 4.1 Affect of demographics on Risk Profile

DO YOU KNOW

AI Algorithms are sometimes accused of suffering of Hungry Judge Effect. The hungry judge effect is a hypothesis that suggests that judges are more likely to make harsher rulings when they are hungry. This effect has been observed in a number of studies, but it is still a subject of debate among researchers.

4.2 ETHICS IN AI

Ethics in AI refers to the principles, guidelines and standards that govern the responsible and ethical development, deployment, and use of artificial intelligence systems. It involves addressing the moral and societal implications of AI technologies to ensure that they are developed and used in ways that align with human values and adhere to ethical norms.

Below we discuss some of the key areas as far as AI in Ethics is concerned:

- **Fairness:** Ensuring that AI systems are unbiased and treat all individuals or groups fairly, without discriminating based on factors such as

race, gender, or socioeconomic status.

- **Transparency:** Promoting transparency in the development and decision-making processes of AI systems, making it clear how algorithms operate and arrive at their conclusions. As highlighted previously, interpretability, explainability and causality are some of the factors in achieving transparency.
- **Accountability:** Establishing mechanisms to assign responsibility when AI systems cause harm or make incorrect decisions and ensuring accountability for developers, organizations and users.
- **Privacy:** Safeguarding individuals' privacy by respecting and protecting their personal data when collecting, processing, or using information within AI systems. This is an issue being looked at very seriously. There is a trade-off here. On one hand, if corporates, governments, NGOs, academic institutes and healthcare providers do not publish open datasets, research will become very difficult, on the other hand, this can raise serious data privacy issues.
- **Security:** Implementing measures to secure AI systems against potential vulnerabilities, hacking, or misuse, preventing unauthorized access or manipulation.
- **Inclusivity:** Considering the diverse perspectives and needs of different stakeholders, ensuring that AI technologies benefit society as a whole rather than exacerbating existing inequalities. There is some overlap in the concept of Inclusivity with that of transparency.
- **Sustainability:** Evaluating the environmental impact of AI systems and working towards minimizing their energy consumption and carbon footprint. AI Models, especially deep learning models have billions of parameters and need as much to be trained on. Hence the computational energy demand is way too high.
- **Robustness and Reliability:** The robustness of an AI model refers to its ability to maintain accurate and reliable performance across diverse and challenging conditions. Achieving robustness is crucial for enhancing the model's resilience and ensuring its reliability in a wide range of practical settings.

Ethics in AI is an evolving field as technology advances and discussions surrounding ethical considerations continue. Many organizations, researchers, and policymakers are actively engaged in developing ethical

frameworks and guidelines to guide the responsible development and use of AI. There is a related concept, which is being discussed called Responsible AI. Issues like robustness and reliability may be more relatable with responsible AI, than Ethical AI.

4.3 BIAS AND FAIRNESS IN AI SYSTEMS

AI seems to be all-powerful in recent days, but what if the models are biased? Bias in AI systems is a critical concern that has garnered significant attention in recent years. It refers to the presence of systematic and unfair discrimination in the outcomes or predictions generated by artificial intelligence algorithms. Bias can manifest in various forms, including but not limited to race, gender, age, socioeconomic status and more. Understanding bias in AI is essential because it has profound implications for individuals, society and the adoption of AI technologies.



DO YOU KNOW?

In 2016, the inaugural international beauty competition, relied on 'machines' for judging the competition. The machine/program was given the responsibility to assess contestants based on objective criteria such as facial symmetry and wrinkles to determine the most attractive participants. Following the launch of *Beauty.AI* earlier that year, over 6,000 individuals from over 100 countries eagerly submitted their photos, anticipating that artificial intelligence, driven by sophisticated algorithms, would identify those whose faces most closely embodied the concept of 'human beauty.' However, upon receiving the results, the organizers were disheartened to discover a troubling commonality among the winners: the robots displayed a clear bias against individuals with darker skin tones.

Let's look at some more historic incidents, which questioned the fairness of the AI Models. The first incident requires a little bit of prelude. Word Embedding is a highly cited work by Google and other teams. It establishes the similarity between two words and this similarity is not confined to trivial examples like linking 'no' with 'not' but can identify that the simi-

larity between man and women is the same as that of king and queen. The word embedding model was doing miracles in terms of relating words. However, in one such example, it is observed that the relationship between ladies and earrings is the same as that of nephews and geniuses. This was taken as sexist, as women were getting related to ornaments and boys with intelligence. In 2016, researchers from Boston University and Microsoft wrote an article titled ‘Man is to Computer Programmer as Woman is to Homemaker’, where they discussed about the somewhat disturbing effects of gender stereotypes.

Another such incident drew quite a lot of attention in 2018. Amazon was using an AI model which was shortlisting candidates for software engineering positions. It was found that the model was penalizing resumes which has mentioned about anything to do with women. You can understand the hue and cry it would have caused. Many of these biases happen because of the training data that they are fed on. In the following section, we discuss about different types of bias.

4.3.1 Types of Bias

Data Bias

Data bias occurs when the training data used to build AI models is unrepresentative or skewed. If the data predominantly contains information from a specific group or exhibits historical prejudices, the AI model can inherit these biases. For example, if a facial recognition dataset contains primarily images of lighter-skinned individuals, the model may perform poorly on darker-skinned individuals. Some more examples of Data Bias are as follows:

- **Gender Bias in Hiring:** If historical hiring data predominantly reflects one gender over another due to past discriminatory practices, algorithms trained on this data may inadvertently favour one gender in current hiring decisions, perpetuating the bias.
- **Racial Bias in Policing:** If police arrest records are used to train predictive policing models, they may result in over-policing in certain neighbourhoods, as arrests in those areas are more frequent, leading

to racial bias in law enforcement. A case regarding recidivism has been discussed already in this context.

- **Age Bias in Loan Approvals:** If a bank's historical data shows a preference for approving loans to younger applicants, AI models trained on this data may discriminate against older applicants seeking loans.
- **Geographical Bias in Healthcare:** Healthcare data may be biased towards urban areas due to better healthcare infrastructure and more comprehensive data collection. This can lead to disparities in healthcare resource allocation in rural or underserved regions.
- **Language Bias in Sentiment Analysis:** Sentiment analysis models trained in a specific language may struggle to understand and interpret sentiments accurately in languages they were not trained in, leading to language bias.
- **Healthcare Diagnosis Bias:** Diagnostic AI systems may provide different accuracy rates across demographic groups due to variations in data quality or representation in the training data, leading to healthcare disparities.

Algorithmic Bias

Algorithmic bias is introduced during the design and development of AI algorithms. It can result from flawed assumptions, model architectures, or feature selections. Biased algorithms may amplify existing prejudices or fail to correct them. For instance, a language translation model might produce gender-biased translations due to gender-neutral languages being misrepresented.

Societal Bias

Societal bias is deeply ingrained in societal values and can influence the decision-making of AI systems. It is often the most challenging bias to identify and mitigate. It goes beyond individual biases and is ingrained in the social, economic and political structures, institutions and norms.

- Societal bias can influence the way data is collected. For example, if there is a bias against a particular racial or ethnic group in society, it may affect the data collected about that group. This can result in the

under representation or misrepresentation of certain groups in datasets.

Schwemmer *et al.,* focussed particularly on detecting gender bias in object recognition systems. They have employed a commercial image recognition system and prepared datasets consisting of US congressmen and public images from the social media platform X (Earlier known as Twitter). Some of the observations from the research article are as follows.

- Three times more annotations related to physical appearance are found in images of women.
- Furthermore, substantially lower recognition rates are observed for women in comparison with men in images.

Figure 4.2 is a revealing finding, where in an exercise, vector embedding of words have been prepared using Wikipedia Corpus. Then similarity of some job roles is calculated with respect to the vectors corresponding to ‘male’ and ‘female’. Occupations like “maestro”, “skipper”, “philosopher“ architect”, “financier”, “pilot” are more similar with the ‘he’ vector (Circles). Occupations like “homemaker”, “nurse”, “receptionist”, “librarian”, “nanny”, “stylist”, “housekeeper”, “counsellor“ are more similar to the she vector (Triangle).

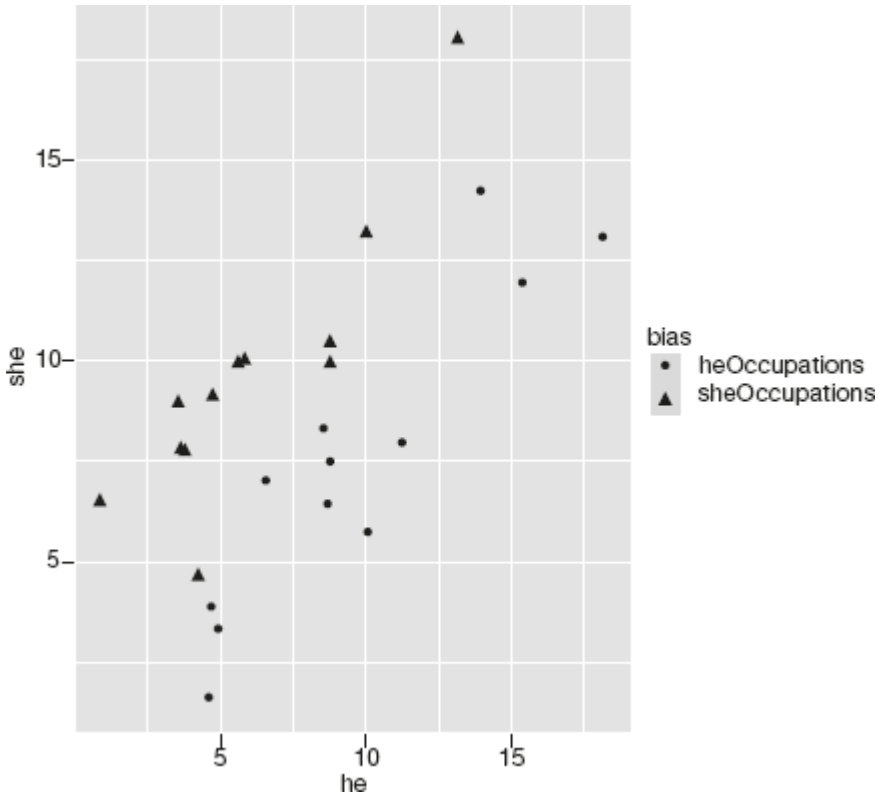


Figure 4.2 Top labels for images of men vs. women from social media

Some immediate finding of gender stereotypes follows:

- More professional work role type words are associated with men, for example, businessperson, spokesperson, and white collar worker while for women it is limited to television presenter.
- Gentleman appears much more formal as compared to 'Girl' as seen in the image labels.
- More physical appearance related to words are present in women.

4.3.2 Measuring and Detecting Bias

One of the fundamental ways of detecting bias is by analysing the data.

Data Review

The first step is to check the data for any variations or differences between groups. If one group is underrepresented or treated unfairly, it might indicate bias. Below are some of the examples,

- In a hiring AI system, if there's a lack of resumes from certain gender groups in the dataset, it indicates data bias. In a loan approval model,

if most of the historical data is from one neighbourhood, it might not be representative of all potential borrowers. If a medical AI system has data only from one region, it may not generalize well to patients from other regions.

Visual Inspection

A picture is worth 1000 words. The same applied to charts. We can use charts and graphs to look for unusual patterns in the data. Odd patterns can suggest bias, especially when comparing different groups.

- Suppose we have a dataset of job applicants, and we want to detect bias in the hiring process. We can create a graph that shows the number of applicants from different racial groups who were hired for the job. If we see that a particular racial group is underrepresented in the hires, this could be a sign of bias in the hiring process. Suppose we have a dataset of student grades, and we want to detect bias in the grading process. We can create a graph that shows the average grade for students from different racial groups. If we see that a particular racial group has a lower average grade, this could be a sign of bias in the grading process. Suppose we have a dataset of criminal arrests, and we want to detect bias in the policing process. We can create a graph that shows the number of people arrested from different racial groups. If we see that a particular racial group is overrepresented in the arrests, this could be a sign of bias in the policing process.
- Essentially the sources of Bias will be similar to those discussed in Data Review. In Visual Inspection, graph-based techniques are employed to find such issues.

Check Fairness

We can calculate fairness metrics related to our specific problem. For example, we can try and see if AI system is making fair decisions for different groups. Fairness metrics are mathematical measures that can be used to assess the fairness of machine learning models. Fairness or parity can be found on many dimensions, one of them being demographic parity.

Let’s find out how we can compute the demographic parity metric:

- **Step 1:** Calculate the prevalence of the positive outcome in each demographic group.
  For example, if we are looking at a loan approval model, we would calculate the percentage of loans that are approved for each demographic group, such as gender, race, and ethnicity.
- **Step 2:** Compare the prevalence of the positive outcome across demographic groups.
  If the prevalence of the positive outcome is equal in all demographic groups, then we say that the model has achieved demographic parity. Otherwise, we say that the model has not achieved demographic parity.

Here is a concrete example:

- Suppose we have a loan approval model that is trained on a dataset of historical loan applications. We want to assess the fairness of the model, so we calculate the demographic parity metric.
- We may find that the prevalence of loan approval is 70% for white males, 65% for white females, 60% for black males, and 55% for black females.
- This means that the model is not achieving demographic parity, because the prevalence of loan approval is not equal across all demographic groups. Black males and black females are less likely to have their loans approved than white males and white females.

Compare Outcomes

It can be investigated if protected groups are affected more than others. If there's a significant difference, it could be a sign of bias.

4.4 TRANSPARENCY IN AI SYSTEMS

As we are increasingly becoming dependent on decisions made by AI, we need to understand the decision-making process of these models to some extent, to have trust in the system. Well, if an oncologist wants to use a computer vision program, which can detect cancer by looking at the MRI Scan, he definitely needs to understand how the model is arriving at this decision. A bank can use an AI-based program for the loan, however, if an

application is rejected based on the same, a higher authority of the bank, or the central bank, or some other regulatory body may want to know the reason for the same.

Transparency in an AI system refers to the clarity and openness in understanding how the system operates, makes decisions and processes information. It involves making the inner workings of the AI system accessible and understandable to users, developers and other stakeholders. Transparency is crucial for building trust, ensuring accountability and addressing concerns related to bias, fairness and ethical considerations in AI.

One thing, that needs to be kept in mind is, as we explore all these concepts related to Ethical AI, like fairness, transparency, accountability, privacy, etc., often these concepts overlap. For example, you can know about the fairness of a model, only when the model is transparent. Let's look at the key aspects related to transparency.

- **Explainability:** The ability to explain how the AI system arrives at a particular decision or recommendation. This involves providing insights into the factors and features considered by the algorithm and the reasons behind its output. Explainable AI (XAI) techniques aim to make complex models more interpretable. For instance, in healthcare, an XAI system might explain the factors considered in predicting a patient's diagnosis, helping physicians and patients comprehend the rationale behind the AI's recommendations and fostering trust in the decision-making process.



DO YOU KNOW?

On March 18, 2018, an Uber self-driving car struck and killed Elaine Herzberg, a pedestrian, in Tempe, Arizona. It was the first known death of a pedestrian caused by a self-driving car. The Uber car was operating in autonomous mode with a human safety backup driver sitting in the driving seat. The safety driver, Rafaela Vasquez, was streaming a television show at the time of the crash. Herzberg was

pushing a bicycle across a four-lane road when the Uber car hit her. The car was traveling at about 40 mph at the time of the impact. Herzberg was taken to the hospital, where she died of her injuries. The National Transportation Safety Board (NTSB) investigated the crash and found that the Uber car's self-driving system was at fault. The NTSB also found that the safety driver's distraction was a contributing factor. In addition to the NTSB investigation, the Arizona Department of Public Safety also conducted a criminal investigation into the crash. In 2019, the Maricopa County Attorney's Office announced that it would not charge Uber with any crimes. However, the office did charge Vasquez with negligent homicide. Vasquez was acquitted of the charge in 2021.

- **Visibility into Data and Training**: Transparency involves disclosing the data used to train the AI system, including information about the sources, quality and representativeness of the data. Knowing the training data helps in understanding potential biases and limitations in the system.
- **Algorithmic Transparency:** Understanding the algorithms and models employed by the AI system is crucial. This includes transparency about the architecture, parameters and optimization methods used in machine learning models. Open-source AI frameworks contribute to algorithmic transparency.
- **Decision-Making Processes:** Transparent AI systems provide clarity on how decisions are made, especially in critical applications like finance, healthcare, or criminal justice. Users should know the criteria considered and the process leading to a specific outcome. Again, decision-making process has some overlap with XAI. The decision-making process looks at the entire sequence of the task, whereas XAI focuses on the model.
- **Ethical Considerations:** Transparency involves addressing ethical considerations in AI development and deployment. This includes being transparent about the ethical principles followed, potential risks and the measures taken to mitigate biases and unfairness.
- **User Interface Design:** A transparent AI system should have an intuitive and user-friendly interface that allows users to interact with and understand the system's functionalities. Clear communication of system capabilities and limitations is essential.

- **Documentation and Reporting:** Providing comprehensive documentation about the AI system's design, functionality and performance metrics contributes to transparency. Regular reporting on the system's performance, updates and any incidents is also important.
- **Compliance with Regulations:** Transparent AI systems adhere to legal and regulatory requirements related to data privacy, security and fairness. Compliance with standards and regulations contributes to accountability and transparency.

While transparency in AI systems is generally considered beneficial, there are some potential negatives to consider.

4.4.1 Security Risks

- **Data Exposure:** Providing too much detail in transparency reports may inadvertently expose sensitive information or vulnerabilities in the system, leading to potential security risks.
- **Model Exploitation:** Detailed insights into the model's architecture and decision-making processes could be exploited by malicious actors to manipulate or attack the system.

4.4.2 Misinterpretation by Users

- **Complexity Overload:** In-depth technical details may overwhelm non-expert users, leading to misinterpretation or misunderstanding of the system's functioning.
- **False Sense of Certainty:** Transparency does not guarantee users' ability to predict every decision accurately and overestimating this capability may result in a false sense of certainty.

4.4.3 Competitive Disadvantage

- **Intellectual Property Concerns:** Full transparency might expose proprietary algorithms or unique features, providing competitors with insights that could impact a company's competitive advantage.
- **Strategic Decision Revealing:** Complete transparency may disclose strategic decisions or business intentions, potentially affecting a company's market position.

It's essential to strike a balance in transparency, providing meaningful insights without compromising security, user comprehension, or a company's competitive edge. Each application and context may require a tailored approach to transparency based on these considerations.

4.5 ACCOUNTABILITY OF THE AI SYSTEMS

Many small and big decisions are being taken by AI Systems, but what happens if something goes wrong? Before proceeding further, let us get introduced to OECD.AI. OECD stands for the Organisation for Economic Co-operation and Development. It is an intergovernmental economic organization with 38 member countries, founded in 1961 to stimulate economic progress and world trade. OECD.AI works under the broad umbrella of the OECD in the area of AI. Some of OECD's observation can be seen below.

The concepts of accountability, responsibility and liability are interconnected yet distinct, and their interpretations may vary across cultures and languages. Broadly, 'accountability' entails an ethical, moral, or similar anticipation, such as those outlined in management practices or codes of conduct, guiding individuals or organizations in their actions. It enables them to elucidate the reasons behind decisions and actions taken. In cases of unfavourable outcomes, accountability involves taking corrective measures to ensure improved results in the future. On the other hand, 'liability' primarily addresses adverse legal consequences resulting from the actions or inaction of an individual or organization.

'Responsibility' also encompasses ethical or moral expectations and finds application in both legal and non-legal contexts to denote a causal connection between an actor and an outcome. Considering these nuances, the term that most aptly captures the underlying principle is 'accountability.' In this context, accountability signifies the expectation that organizations or individuals will guarantee the proper functioning of AI systems they design, develop, operate, or deploy throughout their lifecycle. This alignment should adhere to their respective roles and relevant regulatory frameworks. Demonstrating accountability involves actions and decision-making processes, such as furnishing documentation on pivotal decisions

made during the AI system lifecycle or permitting auditing when warranted.

4.6 PRIVACY AND DATA PROTECTION CONCERNS

Privacy and data protection concerns in the context of artificial intelligence (AI) are critical aspects that demand careful consideration. Let's look at some of the examples where the collectors thought they had taken precautionary measures which proved to be otherwise. The first incident reported was in Massachusetts in 1990. A Government agency was keeping record of all hospital visits. They decided to keep the dataset available for research after duly removing identifying fields like name, address, social security number, etc. They however kept the date of birth, zip code and gender for the sake of summary statistics. Later, a researcher from MIT showed that using this information 87% of the individuals could be identified.

Several AI literatures talk about the Netflix Competition in 2006 as it is a landmark event as far as competitive data science is concerned. This dataset was prepared and shared with the general public after a lot of due diligence in removing all possible identifiable information. Even for this dataset, later some researchers from the University of Texas at Austin could identify some people within just two weeks of publishing the dataset. It may be noted that since 1988, there has been a law in vogue called as Video Privacy Protection Act, which says if any rental company makes the video rental record of any customer public, it is liable for a 2,500 $ fine.

Let's explore the issue of privacy, with the following example of a fictitious hospital in Kolkata:

Name	Age	Gender	Locality	Ailment
Ram	53	M	Maniktala	Heart Disease
Dia	61	F	Salt Lake	Anaemia
Rajen	55	M	Shyam Bazar	Stroke
Anita	48	F	Rajarhat	Typhoid
Nur	62	M	Garia	Dengue
Minati	67	F	Park Circus	Mental Disorder

If now this dataset is released by removing the names, if someone just knows Minati's age and gender, they can find out that she has a mental disorder, which may not be a good thing to happen.

An initial solution of the same is k-anonymity. The idea is even if we can filter by some of the dimensions or attributes, we will not be able to single out individual cases, i.e., there will be at least k individuals satisfying the criteria. For example, instead of the exact age, if we store an age range, there will be two records that are female and have ages between 61–70. Hence, this will be 2-anonymous.

More details of these can be found in the book named 'The Ethical Algorithm'. Another interesting thing is that while there are definitely concerns about sharing the data about let's say, patients or customers, sharing models should not be of any concern at all. Because the models should just expose the architecture and values of different parameters only. However, it turns out that some knowledge of the data also gets leaked through these parameters of the models.

In the same book, the authors have highlighted a very interesting dilemma. Data collection and sharing are necessary for the advancement of knowledge and research, but would anyone be interested in sharing their data, if there is a privacy risk or there is an apprehension that the results may be used against them? Let's say Sunil participates in a survey, which later on establishes a correlation between drinking and increased blood pressure (maybe this survey is a little old when this correlation was not in public knowledge). Now, as this comes to common knowledge and an insurer knows that Sunil drinks, he/she is surely going to charge a hefty insurance premium from Sunil. This argument may put a question mark

on all data sharing. Because the data is anonymized and correlation is established at an aggregate level and still a person like Sunil stands to lose from such a study. An intelligent reader may ask this question, participating in a survey is voluntary. Sunil may not have participated in the survey in the first place. But if we think for a while, we would be conveniently figuring out that, even if Sunil had not participated in the survey, just by having the knowledge that Sunil drinks and drinking causes blood pressure, the insurer any way would have charged a higher premium for his insurance. Frankly, if this is the expectation from AI algorithms that the results will not bring any kind of economic losses, this cannot be ensured. However, what can be ensured is Differential Privacy. Loosely it means, one's differential privacy is preserved, if the inclusion or exclusion of his record in the study, does not change the outcome of the same. More formally it can be stated as follows:

The primary goal of differential privacy is to allow the gathering of useful insights from data while protecting the privacy of individuals in the dataset. It provides a mathematical definition of privacy that quantifies the impact that the inclusion or exclusion of a single individual's data would have on the outcomes of the analysis.

The concept of differential privacy was introduced by a set of theoretical computer scientists in 2000, which actually won them the coveted Godel Prize. One other thing about differential privacy is randomness. When it is used in differential privacy, some amount of noise or disturbance is added to the individual records. This ensures, that the conclusions drawn are not inaccurate as well as the individual records cannot be found out through reverse engineering.

A differential privacy parameter of 1.5, often denoted as 'epsilon' (ε), represents the level of privacy protection in a differentially private system. The epsilon value is a key parameter in differential privacy, and lower values indicate stronger privacy protection.

In the context of differential privacy:

> *A smaller epsilon value corresponds to a higher level of privacy protection. A larger epsilon value allows for more noise or randomness*

to be introduced, potentially offering less privacy protection. The epsilon value is used to control the trade-off between privacy and utility in data analysis. A common interpretation is that smaller epsilon values provide a stricter privacy guarantee but may result in less accurate or useful outcomes.

For example:

- An epsilon value of 0.1 indicates a very strong privacy guarantee, but it may lead to noisier and less accurate results.
- An epsilon value of 1.0 provides a reasonable balance between privacy and utility.
- An epsilon value of 10 allows for more flexibility in the analysis but may provide weaker privacy protection.

It's important to note that the interpretation of epsilon can vary based on the specific differential privacy mechanism and the context in which it is applied. Differentially private algorithms and systems often aim to keep the impact on privacy, measured by epsilon, within a predetermined acceptable range. The choice of epsilon is a critical aspect of designing and implementing differentially private systems to achieve the desired level of privacy while preserving meaningful insights from the data.

Both Apple and Google collect data on usage statistics for malwares. However, they do it by maintaining differential privacy. Let's look at another technique called randomized response. Many times, surveys are on questions which are sensitive or stigmatized. For example, students may be asked whether they are using any substance, employees can be asked if they have taken bribes, etc. The respondents in these cases, will never give correct answers as they know if these responses are leaked, that can cause severe trouble to them. The objective of the surveyor will be defeated here.

So, let's say if the warden of a hostel wants to check on the habit of smoking among the students. The survey can be setup like this:

Do you Smoke?

Randomized Response: Flip a coin. If it is head, then tell the truth, else always say 'you smoke.'

Of course, what the respondent gets in the coin flip is not revealed or recorded. So, if the teacher wants to catch hold of a student and ask him why he or she smokes, he can definitely deny, telling that he/she got a tail, hence he or she said yes.

How does it help the surveyor in getting the percentage of smokers among the students? Well, if it is an unbiased coin, then the probability of head and tail are same i.e., 50%. So, let's assume x% students smoke. 50% of them will get head and will say they smoke, which will be $x/2$%. Rest, all people irrespective of they smoke or not are going to say, they smoke. Hence roughly another 50% will say they smoke. Hence, a total of (50 + $x/2$) % are going to say, they smoke. After the survey, we will know the value of (50 + $x/2$) % and consequently can easily find out what is x. There may be a subtle variation because we assumed exactly 50% heads and tails. However, this should be good enough at an aggregate level.

4.7 SECURITY OF THE AI MODELS

Security concerns regarding an AI model encompass a range of potential risks and vulnerabilities that could impact the confidentiality, integrity and availability of the model and its associated data. As earlier mentioned, not all the issues or concerns that we discuss are isolated. Data security issues will have some overlap with data privacy, transparency, etc.

Some key security considerations include:

- **Making the models safe against adversarial attacks**
  So, what is an adversarial attack?
  An adversarial attack in AI involves manipulating input data to deceive a model's predictions. Imagine you have an image recognition system that's great at identifying objects. An adversarial attack would involve making subtle changes to the input image, almost imperceptible to humans, but causing the AI model to misclassify the object. This type of attack reveals vulnerabilities in the model's decision bound-

aries and prompts efforts to enhance the model's robustness against such manipulations.

Some of the strategies include adversarial training, gradient masking, ensuring input diversity, attack prediction, differential distillation and so on.

- **Making the models immune to deployment risks**

  Deployment typically happens in a different environment, which includes more software components besides the AI model. For access to the models, API (Application Programming Interface) may also need to be exposed. This environment and permissions need to be safeguarded against unauthorized usage and abuse.

  There is a quite famous experiment done by researchers from Carnegie Mellon University, where they showed that the AI-based facial recognition system fails when the person wears a particular type of patterned eyeglasses. These eyeglasses are what we call perturbation to the original pixels. In one of the cases, a white male's picture fooled the AI system into believing it was of a movie actress (**Figure 4.3**).

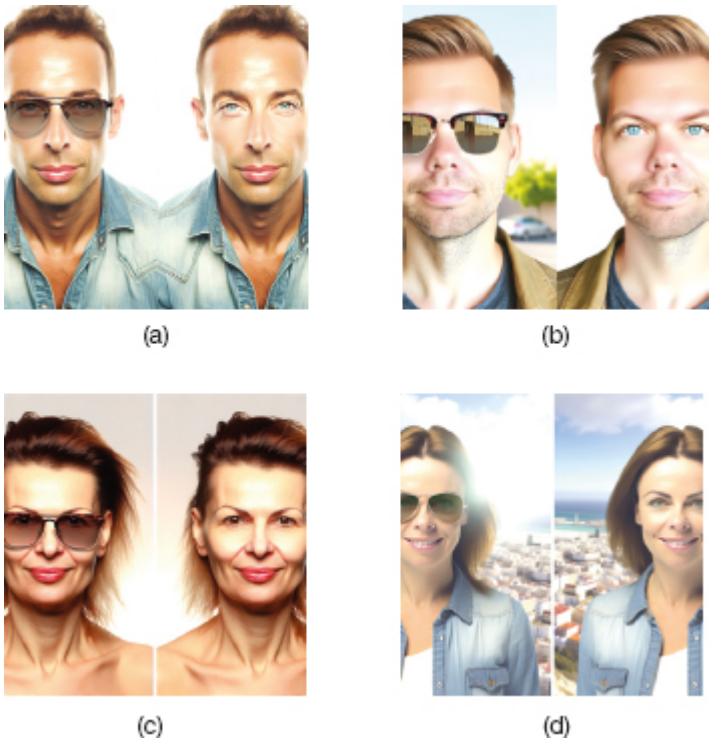


Figure 4.3 Adversarial attack on the images by just adding spectacle

4.8 INCLUSIVITY IN AI

In the context of AI, inclusivity refers to the design and development of artificial intelligence systems in a way that ensures fair and equitable treatment of diverse individuals and groups. It involves creating AI models and technologies that do not discriminate against or exclude people based on factors such as race, gender, age, ethnicity, disability, or other protected characteristics. Again, we would point out there is a decent amount of overlap between fairness and transparency. Hence, we would focus on aspects which plausibly, exclusively map to inclusivity.

- **Accessibility:** Designing AI applications to be accessible to people with diverse abilities and disabilities. This includes considerations for user interfaces, interaction methods and providing alternative means of access.
- **User-Centred Design:** Involving diverse stakeholders, including end-users from various backgrounds, in the design and testing phases to better understand different perspectives and needs.
- **Google Euphonia:** Google Euphonia is one such initiative. It aims to improve speech recognition for diverse populations. It utilizes a massive dataset of speech recordings from speakers with various accents, speaking styles, and disabilities. This data is then used to train AI models to better understand and transcribe speech, regardless of individual differences. This is really a very interesting initiative and interesting readers are welcome to browse more details at (**https://sites.research.google/euphonia/**)
- **SignAll:** This AI-powered app translates sign language into spoken language and vice versa. Again an excellent initiative (**http://www.signall.live/**)

4.9 SUSTAINABILITY IN AI

'Sustainability in AI' refers to the responsible and ethical development, deployment, and use of artificial intelligence technologies to ensure their long-term positive impact on society, the environment and economic systems. We can easily see, again there would be some component which overlaps with ethics and transparency. The most important factor is the environmental impact.

Environmental Impact

Addressing the environmental impact of AI technologies, including the energy consumption of AI models during training and inference. Sustainable AI encourages the development of energy-efficient algorithms and computing infrastructure. The training phase of large and complex neural networks, especially deep learning models, often requires significant computational power and resources. They demand powerful hardware, such as Graphics Processing Units (GPUs) or specialized hardware like TPUs (Tensor Processing Units). The computational intensity of training these models can result in substantial energy usage. They often are run from the data centres which have high energy requirements around the clock. The energy consumed by these AI computations is typically a traditional source of energy, hence increasing carbon footprint. Researchers and engineers are actively working on developing more energy-efficient algorithms and hardware architectures to reduce the environmental impact of AI. Techniques such as model compression, quantization and sparsity optimization aim to achieve similar or improved performance with reduced computational requirements. These are often called ‘Green AI’ practices.

It may be noted that AI can also be used for Sustainability.

As example, it can be used for optimizing resource management in smart grids or precision agriculture. AI can be used for actually most of the Sustainable Development Goals (SDG) as listed by the United Nations (**https://sdgs.un.org/goals**). If we focus on the first SDG that is eradication of poverty. AI has been used to (a) Identify poverty-prone regions; (b) When there are resources available to be distributed, identify the most needy person/household. Similarly, there are many such use cases for the rest of the SDGs.



DO YOU KNOW

‘Distiller’ refers to an open-source library for neural network compression and model optimization in PyTorch. Distiller is designed to facilitate the development and deployment of efficient deep learning models by providing a framework for model compression techniques such as pruning, quantization and knowledge distillation.

4.10 ROBUSTNESS AND RELIABILITY

Robustness in AI models refers to the ability of a model to maintain its performance and generalization capability under various conditions, including situations that differ from those encountered during training. A robust AI model should provide consistent and accurate predictions even when faced with noisy or unexpected input data, changes in the environment, or variations in the distribution of the data. Adversarial attacks, distribution shifts and outliers are common challenges to robustness. Robust models are less sensitive to minor perturbations in input data and can handle a diverse range of scenarios.

Reliability in AI models refers to the consistency and dependability of the model’s predictions over time and across different instances. A reliable AI model produces consistent and trustworthy results. It should demonstrate stability and perform consistently across various inputs and under different conditions. Factors such as data quality, model architecture and training procedures can impact reliability. Reliable models are less prone to producing unexpected or erratic behaviour.

It may be of interest to users to note that, the models often face drifts primarily of two types namely concept drift and data drift.

4.10.1 Concept Drift

Concept drift occurs when the statistical properties of the target variable (the variable you’re trying to predict) change over time. In other words,

the relationship between the input features and the target variable evolves.

Example: If you have a model that predicts stock prices, concept drift may occur if the factors influencing stock prices change over time due to economic conditions, market trends, or other external factors.

4.10.2 Data Drift

Data drift refers to changes in the input features used by a model. It happens when the distribution of the input data changes, but the relationship between input features and the target variable remains constant.

Example: In a spam email classifier, data drift might occur if the characteristics of spam emails change over time, such as a shift in the language used or the types of phishing techniques employed.

4.11 CONCLUSION

AI through its different manifestations have become embedded in our life. For doing a task in lesser time or for a better-quality output we often take the help of AI-based models, algorithms, etc. One of the principal differences between AI-based algorithms with traditional computer science algorithms is these are not explicitly programmed and they learn on their own from the data. Hence, AI needs to be ethical and responsible. Some core issues related to AI ethics are data security and privacy, transparency, accountability, fairness, sustainability, robustness and reliability, etc. Each of these issues is discussed in detail with examples and recent developments.

4.12 SUMMARY

- Over the past two decades, AI programs have achieved significant milestones, from defeating human champions in games like chess and Go to impacting various sectors such as education, healthcare, transportation, and agriculture.
- As AI becomes integrated into daily activities, non-tangible issues like ethics, fairness and bias are emerging concerns, reflecting the need for responsible AI development.

- Ethics, in the context of AI, involves principles and standards guiding professionals in the Information Technology (IT) industry, ensuring morally and socially responsible conduct.
- Issues in AI development include the vast amount of data leading to potential biases, challenges related to data privacy and the lack of complete transparency in the internal workings of AI algorithms.
- The accountability of AI algorithms poses a dilemma, as they may produce unexpected results even with sound design principles, raising questions about transparency, interpretability and causality.
- Visibility into data and training, algorithmic transparency and understanding of decision-making processes are essential for transparency.
- The deployment of AI models, such as self-driving cars and judicial tools like COMPAS, raises ethical concerns, including decision-making in critical situations and potential biases in predicting recidivism.
- Ethics in AI involves principles such as fairness, transparency, accountability, privacy, security, inclusivity, sustainability and robustness, aiming to guide responsible AI development.
- Bias in AI systems is a critical concern, stemming from data bias, algorithmic bias and societal bias, with examples ranging from gender stereotypes in word embeddings to biased hiring practices in Amazon's AI-based recruitment tool.
- Bias detection involves analysing data, reviewing variations between groups and visually inspecting charts for unusual patterns. Fairness metrics, such as demographic parity, can be calculated to assess fairness in AI decision-making.
- Striking a balance is crucial to providing meaningful insights without compromising security, user comprehension, or a company's competitive edge.
- Concerns arise about data sharing's impact on privacy, as illustrated by the concept of Differential Privacy, which aims to protect individuals' privacy in aggregate studies.
- Inclusivity in AI involves fair and equitable treatment, addressing factors such as race, gender, age, and disability.
- Accessibility and user-centric design are essential, ensuring AI applications are usable by diverse populations.
- Sustainability in AI focuses on responsible and ethical development to ensure positive impacts on society, the environment and economic

systems.

- Robust AI models maintain performance under various conditions and are less sensitive to perturbations. Reliability in AI models ensures consistency and dependability over time and across different instances.

4.13 PRACTICE EXERCISES

4.13.1 Subjective Questions (10 Marks)

1. The consideration of Ethics in AI Programs is different from traditional Programs. Justify the statement by enlisting difference between AI Program and traditional Programs.

2. Elaborate on the concept of bias in AI systems. Discuss the different types of bias, such as data bias, algorithmic bias, and societal bias, providing real-world examples for each.

3. Discuss the challenges associated with transparency and explainability in AI algorithms. Why is it important for AI models to be transparent, and what are the implications of a “black box” model?

4. Explain the concept of “differential privacy” and its significance in preserving privacy in AI systems. Provide examples and discuss the trade-offs involved in choosing different epsilon values.

5. Define and discuss the term “accountability” in the context of AI systems. How does accountability differ from liability and responsibility, and why is it crucial for ensuring ethical AI practices?

6. Discuss the significance of visual inspection in detecting bias in AI systems. Provide examples of how graphs and charts can be employed to identify potential bias.

7. Discuss on the overlapping concepts in Ethical AI, such as fairness, transparency, accountability, and privacy. Provide examples of how these concepts intersect.

8. How can fairness metrics contribute to addressing bias in AI systems, and what are their limitations?

4.13.2 Subjective Questions (5 Marks)

1. In the context of AI ethics, discuss the trade-off between the need for open datasets for research and the concerns about data privacy.

2. Explore the potential negative points associated with transparency in AI systems

3. Discuss the challenges and potential risks associated with the security of AI models.

4. Discuss the concept of k-anonymity as a solution and its application in preserving privacy in datasets.

5. Discuss the ethical implications of AI advancements in various sectors such as healthcare, transportation, and education.

6. Explain the process of detecting bias in AI systems using data review and visual inspection techniques. Provide examples to illustrate each step.

7. What are some of the guidelines to make the development and deployment of AI models sustainable.

8. How can AI models be made robust against adversarial attacks, and what measures should be taken to mitigate deployment risks?

4.13.3 Short Answer-type Questions (2 Marks)

1. What is an adversarial attack?

2. Explain what is societal bias?

3. What do you understand by differential privacy?

4. What is algorithmic transparency?

5. What is the primary focus of inclusivity in AI?

6. Define concept drift in context of AI models. Provide examples

7. Define data drift in context of AI models. Provide suitable example

8. What is Green AI ?

9. Explain the difference between robustness and reliability in AI models

4.13.4 Multiple Choice Questions

1. What is the key difference between traditional software and AI software in terms of learning?

a. Traditional software requires explicit programming, while AI software is continuous and learns from data.

b. Traditional software is data-driven, while AI software is rule-based.

c. Traditional software adjusts to new situations, while AI software is inflexible.

d. Traditional software applies to broader problems, while AI software is limited to specific tasks.

2. What is one of the non-tangible issues discussed regarding AI algorithms?

a. Computational efficiency

b. Fairness and bias

c. Legal consequences

d. Rule-based programming

3. What is one of the factors considered in achieving transparency in AI systems?

a. Informed consent

b. Computational efficiency

c. Explainability and interpretability

d. Societal bias

4. What is the key parameter in differential privacy that represents the level of privacy protection?

a. Alpha (α)

b. Beta (β)

c. Delta (δ)

d. Epsilon (ε)

5. In the context of AI models, what does robustness refer to?

a. Consistency and dependability

b. Ability to maintain performance under various conditions

c. Ensuring ethical development

d. Accessibility to diverse individuals

6. What does the term 'accountability' signify in the context of AI systems?

a. Legal consequences

b. Ethical and moral expectations

c. Adherence to regulatory frameworks

d. Anticipation and corrective measures for decisions and actions

7. What does the term ‘responsibility’ encompass in the context of AI systems?

a. Legal consequences

b. Ethical and moral expectations

c. Adherence to regulatory frameworks

d. Expectation of proper functioning throughout the AI system lifecycle

8. What does the term ‘robustness’ refer to in the context of AI models?

a. The ability to maintain accurate and reliable performance across diverse conditions

b. The transparency of the algorithm

c. The continuous learning capability of the AI model

d. The ethical considerations in AI development

9. What is ‘Explainable AI (XAI)’ focused on achieving?

a. Enhancing model resilience

b. Providing comprehensive documentation

c. Making complex models more interpretable

d. Improving transparency in user interface design

10. What should be an ideal first step in detecting bias in AI systems?

a. Check fairness metrics

b. Visual inspection

c. Review the data

d. Compare outcomes

11. What is a potential negative point associated with transparency in AI systems?

a. Enhanced security risks

b. Improved user comprehension

c. Reduced competitive disadvantage

d. Misinterpretation by users

12. How can visual inspection be utilized to detect bias in AI systems?

a. By comparing fairness metrics

b. By analysing the decision-making process

c. By creating graphs and charts to identify the behaviour of the target variable across various groups

d. By calculating transparency scores

4.14 ANSWER KEYS

4.14.1 Multiple-choice Questions

1. (a)

2. (b)

3. (c)

4. (d)

5. (b)

6. (d)

7. (b)

8. (a)

9. (c)

10. (c)

11. (a)

12. (c)